Paper -2(Adv-2022-P2-Model)
FST-06

KEY SHEET

MATHEMATICS

1	**8**	2	**6**	3	**5**	4	**0**	5	**3**	6	**0**
7	**1**	8	**1**	9	**AC**	10	**C**	11	**ABCD**	12	**ACD**
13	**ABC**	14	**BC**	15	**A**	16	**D**	17	**B**	18	**B**

PHYSICS

19	**2**	20	**3**	21	**4**	22	**8**	23	**7**	24	**5**
25	**6**	26	**1**	27	**ACD**	28	**ABC**	29	**ABC**	30	**ABC**
31	**AD**	32	**BCD**	33	**D**	34	**A**	35	**A**	36	**C**

CHEMISTRY

37	**4**	38	**1**	39	**4**	40	**8**	41	**6**	42	**1**
43	**6**	44	**4**	45	**ABC**	46	**ABCD**	47	**ABC**	48	**AB**
49	**ABC**	50	**ACD**	51	**B**	52	**A**	53	**B**	54	**D**

SOLUTIONS
MATHEMATICS

1. $\dfrac{(1+x)^{20}+(1-x)^{20}}{2} = {}^{20}C_0 + {}^{20}C_2x^2 + \ldots {}^{20}C_{20}x^{20} = {}^{20}C_{20} + {}^{20}C_{18}x^2 + \ldots {}^{20}C_0x^{20}$

So, $\sum_{r=0}^{9} {}^{20}C_{2r}\ {}^{20}C_{2r+2} = coeff.\ of\ x^{22}\ in \dfrac{\left[(1+x)^{20}+(1-x)^{20}\right]}{4}$

$\Rightarrow a = 10 \Rightarrow xy = 40$ has total 8 ordered pairs

2. $\sum_{i=1}^{18}(x_i-\alpha)=36, \sum_{i=1}^{18}(x_i-\beta)^2=90 \Rightarrow \sum_{i=1}^{18}x_i = 18(\alpha+2), \sum_{i=1}^{18}(x_i)^2 - 2\beta\sum_{i=1}^{18}(x_i) + 18\beta^2 = 90$

Hence, $\sum_{i=1}^{18}x_i^2 = 90 - 18\beta^2 + 36\beta(\alpha+2)$

Given, $\dfrac{\sum x_i^2}{18} - \left(\dfrac{\sum x_i}{18}\right)^2 = 1 \quad \Rightarrow 90 - 18\beta^2 + 36\beta(\alpha+2) - 18(\alpha+2)^2 = 18$

$\Rightarrow 5 - \beta^2 + 2\alpha\beta + 4\beta - \alpha^2 - 4\alpha - 4 = 1 \Rightarrow (\alpha-\beta)^2 + 4(\alpha-\beta) = 0 \Rightarrow |\alpha-\beta| = 0\ or\ 4$

As α, β are distinct, we get $|\alpha-\beta| = 4$. So, $A = (-\infty,1)\cup(3,\infty)$

$B = (-\infty,-2)\cup(2,\infty)$ $C = (-\infty,2]\cup[6,\infty)$ So, $A\cap B\cap C = (-\infty,-2)\cup[6,\infty)$ and

$(A\cap B\cap C)^c \cap Z = \{-2,-1,0,1,2,3,4,5\}$. Number of subsets will be 256

3. $f(x)+f(1-x)=1 \qquad f\left(\dfrac{1}{40}\right)+f\left(\dfrac{3}{40}\right)+\ldots+f\left(\dfrac{39}{40}\right)=10$

4. $a^2 - b^2 = 0 \therefore$ Number of points = 0 because asymptotes are not tangents

5. $\sum_{n_0=0}^{n_1} 1 = n_1+1,\ \sum_{n_1=0}^{n_2}(n_1+1) = \dfrac{(n_2+1)(n_2+2)}{2!},\ \sum_{n_2=0}^{n_3}\dfrac{(n_2+1)(n_2+2)}{2!} = \dfrac{(n_3+1)(n_3+2)(n_3+3)}{3!}$

So, $\sum_{n_{29}=0}^{n_{30}} \ldots\ldots\ldots \sum_{n_2=0}^{n_3}\sum_{n_1=0}^{n_2}\sum_{n_0=0}^{n_1} 1 = \dfrac{(n_{30}+1)(n_{30}+2)\ldots(n_{30}+30)}{30!}$

And $\sum_{n_{30}=0}^{2}\sum_{n_{29}=0}^{n_{30}}\ldots\sum_{n_2=0}^{n_3}\sum_{n_1=0}^{n_2}\sum_{n_0=0}^{n_1} 1 = \dfrac{(2+1)(2+2)(2+3)\ldots(2+31)}{31!} = \dfrac{33\times32}{2} = 528$

6. $\begin{vmatrix} 2023+2024x^2 & 2023x^2+2024 & 2025 \\ 2017+2018x^2 & 2017x^2+2018 & 2019 \\ 2006+2007x^2 & 2006x^2+2007 & 2008 \end{vmatrix} = 0 \forall x \in R$

7. Dividing given expression with $\cot^2\alpha\cot^2\beta\cot^2\gamma$

$2+\tan^2\alpha+\tan^2\beta+\tan^2\gamma = \tan^2\alpha\tan^2\beta\tan^2\gamma$

$2+\cos ec^2\alpha-1+\cos ec^2\beta-1+\cos ec\gamma-1=\left(\cos ec^2\alpha-1\right)\left(\cos ec^2\beta-1\right)\left(\cos ec^2\gamma-1\right)$

$\Rightarrow \sec^2\alpha\sec^2\beta+\sec^2\beta\sec^2\gamma+\sec^2\gamma\sec^2\alpha=\sec^2\alpha\sec^2\beta\sec^2\gamma$

$\Rightarrow \dfrac{1}{\sec^2\alpha}+\dfrac{1}{\sec^2\beta}+\dfrac{1}{\sec^2\gamma}=1 \;\; \cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

8. $(x+2y+1)dy-(2x-y+1)dx=0 \quad \Rightarrow 2(ydy-xdx)+(xdy+ydx)+(dy-dx)=0$

$\Rightarrow y^2-x^2+xy+y-x=c \qquad y(1)=1\Rightarrow c=1$

So, solution will be $y^2-x^2+xy+y-x=1 \;\; x=0\Rightarrow y^2+y-1=0$

9. Given, the vertices of the triangle are $O(0,0), A(a,0), B(0,b)$

Let $C(h,k)$ be the centre and r be the radius of the given circle

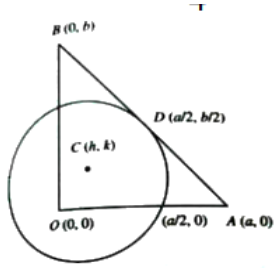
Since the circle touches AB at point $D\left(\dfrac{a}{2},\dfrac{b}{2}\right)$, the centre $C(h,k)$ lies on the line

$y-\dfrac{b}{2}=\dfrac{a}{b}\left(x-\dfrac{a}{2}\right)$ i.e., $2ax-2by=a^2-b^2$

So, $2ah-2bk=a^2-b^2$ Also, $r^2=CD^2=\left(h-\dfrac{a}{2}\right)^2+\left(k-\dfrac{b}{2}\right)^2$

And also $r^2=\left(h-\dfrac{a}{2}\right)^2+k^2$ because the circle passes through $\left(\dfrac{a}{2},0\right)$, the midpoint of

the smaller side, Solving the above equations, we get $h=\dfrac{2a^2-b^2}{4a}$ and $r=\dfrac{b\sqrt{a^2+b^2}}{4a}$

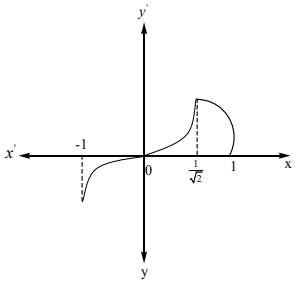


10. Conceptual

11. $\alpha+\beta^2=4, \alpha\beta^2=\dfrac{3k}{5}, \beta\alpha^2=\dfrac{45}{k}, \beta+\alpha^2=\dfrac{50}{k}\Rightarrow \alpha^3\beta^3=27\Rightarrow \alpha\beta=3$

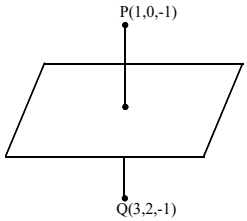
$\Rightarrow \alpha=3,\beta=1,k=5$

12.



$$h(x)=\begin{cases}\sin^{-1}x & for\;\; -1\le x<\dfrac{1}{\sqrt{2}}\\ \cos^{-1}x & for\;\; \dfrac{1}{\sqrt{2}}\le x\le 1\end{cases}$$

13.



R is the midpoint of PQ So, $R(2,1,-1)$ lie on plane, Direction ratios of the normal to the plane will be $<1,1,0>$, which means the plane equation will be

$1(x-2)+1(y-1)=0 \Rightarrow x+y=3$

14. $$f(\theta)=\sum_{n=1}^{\infty}\frac{\tan\frac{\theta}{2^n}}{2^{n-1}\cos\frac{\theta}{2^{n-1}}}=\sum_{n=1}^{\infty}\frac{\sin\frac{\theta}{2^n}}{2^{n-1}\cos\frac{\theta}{2^n}\cos\frac{\theta}{2^{n-1}}}$$

$$f(\theta)=\sum_{n=1}^{\infty}\left(\frac{1}{2^{n-2}\sin\frac{\theta}{2^{n-2}}}-\frac{1}{2^{n-1}\sin\frac{\theta}{2^{n-1}}}\right)=\frac{2}{\sin 2\theta}-\frac{1}{\theta}$$

15. The class interval containing $\frac{N^{th}}{2}$ or 25th item is 20-30. Therefore, 20-30 is the median class. We know that , Median $=l+\frac{\frac{N}{2}-C}{f}\times h$, Here $l=20,\ C=13,\ f=15,\ h=10$ and $N=50$, Therefore, Median $=20+\frac{25-13}{15}\times 10=20+8=28$

Thus, Mean deviation about median is given by

M.D. (M) $=\frac{1}{N}\sum_{i=1}^{6}f_i\left|x_i-M\right|=\frac{1}{50}\times 508=10.16$

16. $A=\begin{bmatrix}3 & 1\\ -6 & -2\end{bmatrix}\Rightarrow |A|=0$, So, A satisfy $A^2-A=O$

$(I+A)^{2025}=I+\left({}^{2025}C_1A+{}^{2025}C_2A+...+{}^{2025}C_{2025}A\right)$

17. $$\int\frac{\sec^2 x}{(\sin x)^{2025}}dx-\int\frac{2025}{(\sin x)^{2025}}dx=\frac{\tan x}{(\sin x)^{2025}}+2025\int\frac{\tan x\cos x}{(\sin x)^{2026}}dx-\int\frac{2025}{(\sin x)^{2025}}dx=\frac{\tan x}{(\sin x)^{2025}}+C$$

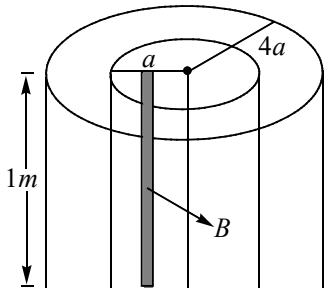
18. If $n=m$, then the equation becomes $z^m=(\bar{z})^m$, which means it has infinite solutions because z gets satisfied by all real numbers, If $n\neq m$, then

$z^n=(\bar{z})^m\Rightarrow |z|^n=|\bar{z}|^m\Rightarrow |z|=1$ or 0

So, number of solutions will $(n+m+1)$

PHYSICS

19.

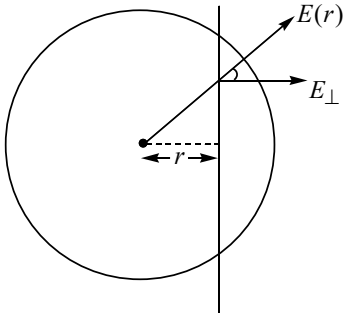


$$\int_0^a \left(\frac{\mu 0 Jr}{2}\right)^2 \frac{\mu\pi r dr}{2\mu} + \int_\alpha^{4a} \frac{1}{2}\left(\frac{\mu_0 J a^2}{2r}\right)\frac{1}{\mu_0} 2\pi r dr$$

$$\Rightarrow \frac{\mu_0 J^2\pi}{4}\frac{a^2}{4} + \frac{\mu_0 J^2 a^4 \pi}{4}\ln 4 = \frac{1}{2} L J^2 \pi^2 a^4$$

$$\Rightarrow \frac{\mu_0}{\pi}\left[\frac{1}{8} + \frac{\ln 4}{2}\right] = L \qquad \Rightarrow \frac{\mu_0}{\pi}[0.82] = L$$

20.



$$E(r) = \frac{\rho r}{3\in_0}, E\perp = \frac{\rho r\cos\theta}{3\in_0} = \frac{\rho x}{3\in_0} \quad \phi = \frac{\rho x}{3\in_0}\times\pi\left(R^2 - x^2\right) \qquad = \frac{\rho\pi}{3\in_0}\left(R_x^2 - x^3\right)$$

For maximum $\phi, \frac{d\theta}{dx} = 0,$ $\qquad R^2 - 3x^2 = 0 \rightarrow x = \frac{R}{\sqrt{3}}$

So, $\phi = \frac{\rho\pi}{3\in_0}\left(R^2\frac{R}{\sqrt{3}} - \frac{R^3}{3\sqrt{3}}\right)$

Putting $R = \sqrt{3}$ $\qquad \phi_{\max} = 2\frac{\rho\pi}{3\in_0} = \frac{\rho\pi}{\left(\frac{3}{2}\right)\in_0}$ N = 1.5

21. For convex lens $\qquad v_1 = \frac{-36f}{-36+f}$

Distance of image from plain mirror

$$d = \frac{36f}{-f+36} - 100 = \frac{36f + 100 - 3600}{36 - f} \qquad = \frac{136f - 3600}{36 - f}$$

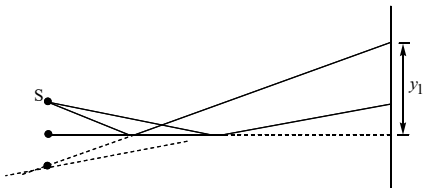
For concave lens

$$u = \frac{136f - 3600}{36 - f} - 65 = \frac{201f - 5940}{36 - f}$$

$$\frac{1}{20}-\frac{36-f}{201f-5940}=\frac{1}{-2f} \qquad \Rightarrow f=30\ cm$$

Now from lens maker formula $\mu_1=2.5$ and $\mu_2=1.5$

22.



$$\beta=\frac{\lambda D}{d}=\frac{500\times10^{-9}\times1}{1\times10^{-3}}=500\times10^{-6}=0.5\ mm$$

$$\frac{y_1}{90}=\frac{0.5mm}{10} \qquad y_1=4.5\ mm$$

For maxima $y\dfrac{d}{D}=(2N+1)\dfrac{\lambda}{2}$

$$y=(2N+1)\frac{\lambda D}{2d} \quad \frac{\lambda D}{d}=0.5\ mm \quad y=\frac{(2N+1)(0.5\ mm)}{2}$$

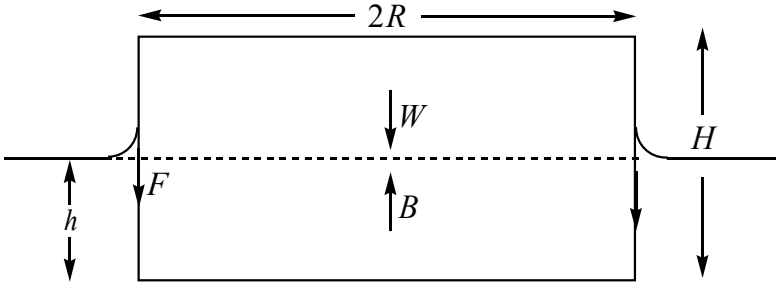
$$2N+1=17 \quad y=\left(\frac{17}{2}\right)\left(0.5\ mm\right)=(8.5)(0.5) \quad =4.25\ mm$$

23. $v_0=\sqrt{\dfrac{2GM}{R}}$ By energy conservation $\dfrac{1}{2}mv_0^2-\dfrac{GMm}{R}=\dfrac{1}{2}mv^2-\dfrac{GMm}{r}$

$$v=\sqrt{\frac{2GM}{r}} \quad \int_{r=R}^{r=4R}\sqrt{r}\ dr=\sqrt{2GM}\int_0^t dt \quad t=\frac{7}{3}\sqrt{\frac{2R^3}{GM}}$$

24. $\dfrac{E_0 d}{e\ell}=V$

25.



The disk is in equilibrium under forces of gravity, buoyant force and surface tension.

W + F = B $\quad \pi R^2 H 0.5\rho_w g+S.2\pi R=\pi R^2 h\rho_w g$

$$14\times14\times10\times10^{-9}\times0.5\times10^3\times10+0.07\times2\times14\times10^{-3}=14\times14\times h\times10^{-9}\times10^3\times10$$

$$98\times10^{-4}+1.96\times10^{-3}=196\times10^{-5}h$$

$$98\times10^{-4}+19.6\times10^{-4}=19.6\times10^{-4}h \qquad \text{H = 6 mm}$$

26. $a=\dfrac{v_2-v_1}{t}=\dfrac{0.85-0.21}{8.0}=0.08\,m/s^2 \quad \dfrac{\Delta a}{a}=\dfrac{\Delta(v_2-v_1)}{(v_2-v_1)}+\dfrac{\Delta t}{t}=\dfrac{\Delta v_2+\Delta v_1}{v_2-v_1}+\dfrac{\Delta t}{t}$

$$\frac{\Delta a}{a}=\frac{0.05+0.05}{0.85-0.21}+\frac{0.1}{8.0} \quad =\frac{0.1}{0.64}+\frac{0.1}{8.0}$$

$$\Delta a=0.08\left(\frac{0.1}{0.64}+\frac{0.1}{8.0}\right)=0.0125+0.001 \qquad =0.0135$$

Since error is in the 2nd place of decimal so Δa must be rounded to 0.0. So acceleration will be quested as $0.08\pm0.01\ |a|+|\Delta a|=0.09$

27. A) Any line passing through origin in a T-V graph represents constant pressure. The slope of this line is $\frac{T}{V}$ which is directly proportional to pressure P.

During process DA the slope of line through origin increases in slope indicting increase in pressure. Also the line joining origin and point C also intersects DA at a point.

B) $Q_{AB}=nC_p\Delta T=\frac{5}{2}R(T_B-T_0)=\frac{5}{2}T_0 \quad Q_{BC}=nRT\ln\frac{V_C}{V_B}=R2T_0\ln 2$

$$Q_{CD}=nC_v\Delta T=-\frac{3R}{2}T_0 \quad Q_{DA}=-RT_0\ln 4=-2RT_0\ln 2$$

C) $\eta=\dfrac{\text{work done by gas}}{\text{Heat absorbed}}$

$$W_{cycle}=W_{AB}+W_{BC}+W_{CD}+W_{DA}=RT_0+2RT_0\ln 2+0-2RT_0\ln 2=RT_0$$

$$\eta=\frac{RT_0}{\frac{5}{2}RT_0+2RT_0\ln 2}=\frac{2}{5+4\ln 2}$$

D) Maximum slope of T-V line is for process AB

$$P=nR\frac{T}{V} \quad P_{\max}=\frac{RT_0}{V_0}$$

28. Ceq (before any attempt) $=\frac{2}{5}C \qquad Ceq_1=\frac{4}{9}C$ and $Ceq_2=\frac{6}{11}C$

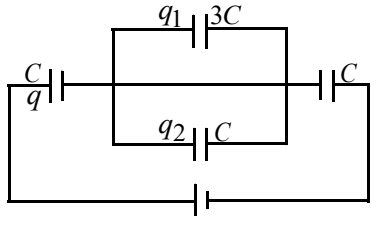
$$\Delta C_1=1\text{st change}=\frac{4}{9}C-\frac{2}{5}C=\frac{2}{45}C \quad \Delta C_2=2\text{nd chance}=\frac{6}{11}C-\frac{2C}{5}=\frac{8}{55}C$$

A) $\frac{2}{45}C:\frac{8}{55}C=\frac{11}{36}$

B) $W_{cell}=\Delta_q\times emf=\Delta C\times\varepsilon^2$

So ratio of work done is same as ratio of option (A)

C) After attempt-I circuit becomes



$$q_1 = \frac{3}{4}q \quad \frac{U_A}{U_B} = \frac{q^2}{2C} \Bigg/ \frac{\left(\frac{3}{4}q\right)^2}{6C} = 16:3$$

D) Heat dissipated cannot be calculated as external work is involved

29. The ring in horizontal plane is cutting the flux to time varying component of $\vec{B}$

emf $\varepsilon = \pi R^2 \frac{d}{dt}\left(B_0 t^2\right) = 2\pi R^2 B_0 t \qquad i = \frac{2\pi R^2 B_0 t}{\lambda 2\pi R} = \frac{RB_0 t}{\lambda}$

'M' magnetic moment due to this current $\pi R^2 \frac{RB_0 t}{\lambda}$. The direction of M is negative y according to Renz's law.

This is perpendicular to $t_0 B_0 i$, so a torque starts acting having direction $-j \times i = k$.

This torque is balanced by the supports so reaction at S_1 increases and S_2 decreases (B).

Also net force is only gravitational so (A).

Let reaction at S_1 and S_2 be N_1 & N_2

$$(N_1 - N_2)R = \tau_B = \frac{\pi R^3 B_0^2 t}{\lambda} \qquad N_1 - N_2 = \frac{\pi R^2 B_0^2 t}{\lambda}(C)$$

It looses constant from S_2 when torque of gravity balances the magnetic torque about S_1

$$2mgR = \frac{\pi R^3 B_0^2 t}{\lambda} \rightarrow t = \frac{2mg\lambda}{B_0^2 \pi R^2}$$

30. Electric field in the cavity is given by $\vec{E} = \frac{\rho \vec{b}}{3 \in_0}$ where $\vec{b}$ is the vector from center of outer sphere center of cavity.

Here $E = \rho \frac{(1-\eta)R}{3 \in_0}$ for $\eta \rightarrow 0, E = \frac{\rho R}{3 \in_0}$

for $\eta = \frac{1}{2}, E = \frac{\rho R}{6 \in_0}$ directed along BD

So. $2\eta R = \frac{1}{2}at^2$, where $a = \frac{qE}{m}$ $t = \sqrt{\frac{12 \in_0 m}{\rho q}}$

(C) It strikes at D

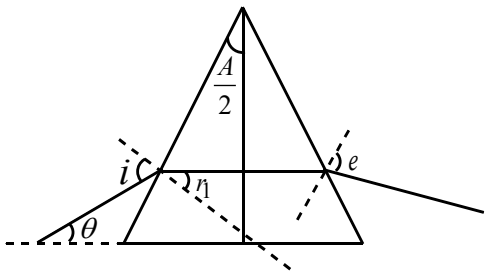
Along y $\qquad t = \frac{\eta R}{v_1}$

Along x

$$\frac{1}{2}at^2 = \eta R \rightarrow \frac{1}{2}\rho \frac{R}{6 \in_0}\frac{q}{m}\frac{\eta^2 R^2}{v_1^2} = \eta R \quad v_1 = \frac{R}{2}\sqrt{\frac{\rho q}{6 \in_0 m}}$$

D) $\Delta K.E = W = qE\frac{R}{2} = q\frac{\rho R}{6 \in_0}\frac{R}{2} = \frac{\rho q R^2}{12 \in_0}$

31.



From geometry $i = \theta + \frac{A}{2}$

For Prism-1 $i = 15^\circ + 30^\circ = 45^\circ$

By snell's Law at 1st surface

$\sin 45^\circ = \sqrt{2}\sin r_1,\ r_1 = 30^\circ \rightarrow r_2 = 30^\circ$

So $e = 45^\circ \quad \delta = i + e - A = 30^\circ$

As incident ray is same and emergent rays are parallel this suggests that for each prism condition for minimum deviation is satisfied.

for Prism-2 $i = 15^\circ + 15^\circ = 30^\circ$ and $r = \frac{A}{2} = 15^\circ$

So $n^2 = \frac{\sin 30^\circ}{\sin 15^\circ} = 2\cos 15^\circ = 2\cos\left(\frac{\pi}{24}\right)$

for prism-3 $i = 15^\circ + 45^\circ = 60^\circ$ and $r = \frac{A}{2} = 30^\circ$

So $n_3 = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{\frac{3}{2}}$

32. Let mass of rim be equal to m

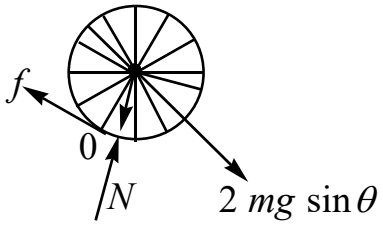
Applying NLM to wheel

$2\,mg\sin\theta - f = 2\,ma$ .....(1)

In the C frame of wheel

$\tau = I\alpha \qquad fR = \left(mR^2 + \frac{mR^2}{3}\right)\alpha \quad f = \frac{4}{3}ma$

Solving (1) and (2)



$a = \frac{3}{5}g\sin\theta \rightarrow f = \frac{4}{3}\times\frac{3}{5}mg\sin\theta = \frac{4}{5}mg\sin\theta$

But $f \le \mu N \rightarrow \frac{4}{5}mg\sin\alpha \le \mu 2mg\cos\theta \rightarrow \mu \ge \frac{2}{5}\tan\theta$

At a given moment all spokes are in pure rotation about point 'O' of contact. So ratio of KE will be the ratio of moment of inertia about 'O'.

$$KE_L : kE_H = \frac{mR^2}{3} : \left(\frac{mR^2}{12} + m\left(\frac{3R}{2}\right)^2 \right) = 1:7$$

$$\frac{dL}{dt} = \tau = \tau_g = 5g\sin\theta \times R = 5 \times 10 \times \frac{1}{2} \times 0.4 = 10\, kg\, m^2 s^{-1}$$

33. $\frac{Q_{out}}{Q_{in}} = \frac{T_{out}}{T_{in}}$ and $Q_{out} = Q_{in} + W \qquad \Rightarrow Q_{in} = W\left(\frac{T_L}{T_H - T_L}\right)$

$$\frac{Q_{in}}{t} = P\left(\frac{T_L}{T_H - T_L}\right)$$

34. Between faces LMNO and PQRS

$$dR = \frac{\rho dr}{t\frac{\pi}{2}r} \qquad R = \frac{2\rho}{t\pi}\ell n\frac{b}{a}$$

Between faces LMRS and ONQP for an element at distance r from O. $dR = \frac{\rho\frac{\pi}{2}r}{tdr}$

So, $\frac{1}{R} = \int\frac{1}{dR} = \int_a^b \frac{tdr}{\rho\frac{\pi}{2}r} = \frac{2t}{\rho\pi}\ell n\left(\frac{b}{a}\right) \qquad R = \frac{\rho\pi}{2t\ell n\left(\frac{b}{a}\right)}$

35. $v_0 = 4\sqrt{gR} \quad \vec{J} = \Delta\vec{P}$

$$-\int Tdt = 2M\left(v - 4\sqrt{gR}\right)$$

$$-\int Tdt = 2M\left(R\omega - 4\sqrt{gR}\right) \quad .....(i)$$

$$\left(\int Tdr\right)R = MR^2\omega \quad .....(ii)$$

From (i) and (ii)

$$\omega = \frac{8}{3}\sqrt{\frac{g}{R}} \text{ and } v = \frac{8}{3}\sqrt{gR}$$

36. $x_{cm} = \frac{\int xdm}{\int dm} = \frac{\int_0^L x\lambda_0\left(1+\frac{x}{L}\right)}{\int_0^L \lambda_0\left(1+\frac{x}{L}\right)} = \frac{5L}{9}$

$$m = \int dm = \frac{3\lambda_0 L}{2}$$

$$\left[\left(A.\frac{L}{9}\right)\rho g\right]\frac{17}{18}L\sin\theta = F.L\cos\theta$$

CHEMISTRY

37. $C_{vm} = 3R \quad C_{pm} = 4R \quad y = \frac{4}{3} \qquad TV^{y-1} = K$

$T_2(2)^{y-1} = T_1(16)^{y-1} \qquad \frac{T_2}{T_1} = (8)^{y-1} \quad \frac{T_2}{T_1} = 2$

Let $T_1 = T$ then $T_2 = 2T \qquad n = \frac{T_2 - T_1}{T_2} = 0.5 \qquad n = \frac{-W_{total}}{Q_2}$

$0.5 = \frac{-W_{total}}{8} \qquad W_{total} = -4\,k\,cal \qquad |W_{total}| = 4\,k.cal$

38. The relevant equilibrium with problem and the overall equilibrium is as given below.

$Agl_{(s)} \rightleftharpoons Ag^+ + I^- \qquad K_{SP} = 1.2 \times 10^{-7}\ M^2$

$Ag^+ + 2CN^- \rightleftharpoons [Ag(CN)_2]\ K_1 = 7.1 \times 10^{19}\ M^{-2}$

$Agl_{(s)} + 2CN^- \rightleftharpoons [Ag(CN)_2]^- + I^- \quad K = 1.2 \times 7.1 \times 10^{12}$

Initially	1M	0	0	$= 8.52 \times 10^{12}$
At equilibrium	$(1-2x)M$	xM	xM	

Where x is the solubility of Agl in mole L^{-1}

$\frac{x^2}{(1-2x)^2} = 8.52 \times 10^{12} \qquad \Rightarrow \frac{x}{1-2x} = 2.91 \times 10^6$

$\Rightarrow x = 2.91 \times 10^6 - 5.82 \times 10^6 x \quad \Rightarrow x + 5.82 \times 10^6 x = 2.91 \times 10^6$

$\Rightarrow 5.82 \times 10^6 x = 2.91 \times 10^6 \qquad \Rightarrow 5.82 \times 10^6 x = 2.91 \times 10^6 \Rightarrow x = \frac{2.91 \times 10^6}{5.82 \times 10^6} = 0.5$

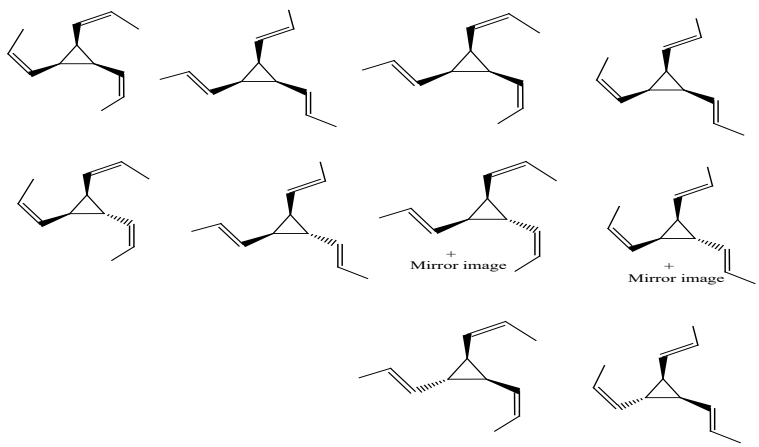
39. $Z = 1 + \frac{Pb}{RT} \quad \frac{dz}{dp} = \frac{b}{RT} \qquad 0.01 = \frac{b}{RT}$ B = 0.01 RT

Y-intercept = nRT 40 = 2RT , RT = 20 $b = 0.01 \times 20 \qquad = 0.2$

Excluded volume for 20 moles of the real gas = 20 b = 4

40. X = 5, y = 3

41.



42. $C_2 : 12e^- : \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2$

$O_2^- : 17e^- : \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2 \pi^* 2p_x^2 \pi^* 2p_y^1$

43. $\left[Cr^{III}\left(NH_3\right)_4 BrCl\right]Cl : Cr^{3+} : 3d^3 : t_{2g}^3 e_g^0$

$\left[Cr^{III}\left(NH_3\right)_4 Cl_2\right]Br : Cr^{3+} : 3d^3 : t_{2g}^3 e_g^0$

44. $3NiS + 2HNO_3 + 6HCl \rightarrow 2NO + 3NiCl_2 + 3S + 4H_2O$

$CoS + HNO_3 + 3HCl \rightarrow CoCl_2 + NOCl + S + 2H_2O$

45. Conceptual

46. Conceptual

47. (D $NaBH_4$ CAN NOT REDUCE ESTERS)

48. Conceptual

49. Conceptual

50. MnO_4^{2-} : coloured and paramagnetic

51. $\frac{(B)}{(C)} = \frac{2K_1}{3K_2} = \frac{4}{9}$

(B) + (C) = 2

$(B) + \frac{9(B)}{4} = 2$

$\frac{13(B)}{4} = 2$

$(B) = \frac{8}{13}$

$(C) = \frac{9(B)}{4} = \frac{18}{13}$

52. Option A requires 3 times Hoffmann elemination to remove the nitrogen from the ring

53. Michael addition followed by Aldol condensation

54. In a cyclotrimetaphosphoric acid 3 double bonds and 12 single bonds are present.